Physics-Aware Multi-Objective Scheduling for an Agile SAR Satellite: Balancing Profit, NESZ, and Geometric Resolution

LIU Shuhao

College of Intelligent Systems Science and Engineering, Harbin Engineering University, 145 Nantong Street, Nangang District, Harbin, 150001, China

Abstract

In agile SAR scheduling, observation timing determines not only feasibility but also imaging quality through radar-specific geometric and radiometric effects. We characterize an empirical saturation regime for multi-objective SAR quality optimization in single-pass, single-satellite Stripmap operation by integrating two physical quality dimensions, Noise-Equivalent Sigma Zero (NESZ, radiometric) and geometric resolution, as decomposed optimization objectives within a Non-dominated Sorting Genetic Algorithm III (NSGA-III) framework, distinct from scalar visibility composites used in prior SAR scheduling. Experiments spanning target densities from 20 to 500 show a scale-dependent saturation: at sparse densities ($N \leq 100$), multi-objective optimization can extract modest geometric quality gains (up to +8.2% in f_2 at the sparsest density, $N = 20$) at a steep coverage cost (roughly one third of f_1^*), while at operational densities ($N \geq 300$), multi-objective evolutionary algorithm (MOEA) formulations seeded by the greedy baseline (G-BL) approach the greedy coverage anchor and the marginal quality gain falls below 2%. This coverage convergence is conditional on greedy-baseline hot-start initialization, with random-init f_1^* deficits of 0.18–0.46 at all densities. Independently, ablation shows that removing the physical objectives barely changes dense-regime outcomes. The additional third objective is effectively redundant: MOEA-3 yields results practically indistinguishable from MOEA-2, reflecting the geometric coupling of the low-squint regime rather than a

Email address: `liushuhao@hrbeu.edu.cn` (LIU Shuhao)

mathematical identity between objectives. Rather than identifying a superior solver, our results establish an empirical saturation boundary for physics-aware multi-objective scheduling under the tested single-satellite, single-pass, Stripmap, G-BL-seeded NSGA-III conditions.

Keywords: agile SAR, satellite scheduling, multi-objective optimization, NSGA-III, noise-equivalent sigma zero (NESZ), geometric resolution

---

1. Introduction

Agile Earth-observation satellites equipped with Synthetic Aperture Radar (SAR) are increasingly central to time-critical monitoring missions, from disaster response to maritime surveillance. Their agility (the ability to slew the antenna beam away from the nadir track) enables a single satellite to image far more targets per orbit than a fixed-pointing platform (Curlander and McDonough, 1991). This flexibility, however, transforms scheduling from a simple feasibility check into a combinatorial optimization problem: for N targets each with a finite visibility window, the scheduler must select a subset and assign observation times that satisfy attitude transition constraints while maximizing some measure of mission value. In conventional satellite scheduling, observation quality is commonly represented as a fixed attribute of each pre-computed opportunity rather than a quantity that varies with the selected acquisition time. The scheduling problem therefore reduces to maximizing the number or priority-weighted sum of observed targets subject to temporal feasibility. This assumption underpins most existing satellite scheduling work, from mixed-integer programming formulations to deep reinforcement learning approaches (Chen et al., 2025).

SAR imaging inherently departs from this assumption because image quality depends on the radar geometry at the moment of acquisition: the ground-range resolution degrades with incidence angle ($\rho_{\text{rg}} \propto 1/\sin\theta$), the azimuth resolution broadens with squint angle ($\rho_{\text{az}} \propto 1/\cos\psi_{\text{sq}}$), and the radiometric sensitivity (Noise-Equivalent Sigma Zero, NESZ) deteriorates with slant range as R^3 (Cumming and Wong, 2005). When the scheduler can choose *when* within a visibility window to observe each target (the defining feature of agile satellites), it simultaneously chooses the imaging quality at which each target is acquired. Despite this tight coupling between scheduling and imaging physics, with few exceptions (Wei et al., 2025; Kramer and Miller, 2025), existing agile SAR scheduling methods either reduce SAR to a coverage-only objective or apply post hoc quality checks on schedules produced by coverage-maximizing solvers. Few approaches treat physical quality dimensions as primary optimization objectives in the SAR domain.

The problem is NP-hard by reduction from the Orienteering Problem with Time-Dependent Profits (Peng et al., 2019), precluding polynomial-time exact algorithms for all but the smallest instances. The SAR literature has well-established models for NESZ and geometric resolution as functions of observation geometry (Curlander and McDonough, 1991; Cumming and Wong, 2005), and the multi-objective optimization literature provides mature frameworks such as the Non-dominated Sorting Genetic Algorithm III (NSGA-III) (Deb and Jain, 2014) for handling three or more objectives simultaneously. Yet direct integration of these three elements (SAR physical modeling, multi-objective evolutionary algorithms (MOEAs), and satellite scheduling) remains limited. We therefore formulate SAR scheduling with separate, physics-derived geometric and radiometric quality objectives within a MOEA framework.

We formulate a three-objective agile SAR scheduling problem in which the scheduler optimizes (1) total coverage profit, (2) mean geometric image quality (ground-range and azimuth resolution), and (3) mean NESZ radiometric quality simultaneously. We compare five solvers spanning the methodological spectrum: two greedy heuristics (one profit-only baseline and one squint-minimized variant), a single-objective genetic algorithm with greedy hot-start, and two NSGA-III variants with two and three objectives. The solvers are evaluated on 200 systematically varied scenarios across four density classes ($N = 20$ to $N = 500$). Our results show a scale-dependent saturation, not a uniform solver advantage. In sparse scenarios, multi-objective optimization can extract modest geometric quality gains, but at a steep coverage cost, selecting only a fraction of the targets captured by a simple profit-only greedy scheduler. In dense scenarios, the multi-objective advantage diminishes to negligible levels: greedy methods approach Pareto-optimal coverage, and the squint-minimized heuristic offers competitive radiometric quality. Section 7.1 provides a detailed synthesis of these findings. The specific contributions of the paper are:

1. The discovery and analytical characterization of an empirical saturation regime for multi-objective SAR quality optimization in the single-pass, single-satellite Stripmap setting: the regime arises from solver convergence to low-squint regimes that suppress the θ-axis conflict between geometric and radiometric quality (§ 6.4), not from a physical

identity between objectives, and renders the three-objective formulation redundant at operational densities. A controlled ablation study (four objective-function variants, 200 scenarios per variant) confirms that solver outcomes become invariant to the choice of physical objectives at high target densities, and initialization controls (§ 6.6) confirm that coverage convergence is conditional on greedy-baseline (G-BL) hot-start initialization;

2. The decomposition of SAR image quality into closed-form NESZ ($\propto R^3$) and geometric quality index ($\sin\theta\cos\psi_{\mathrm{sq}}$) objectives within an NSGA-III framework, distinct from the scalar visibility composite of Kramer and Miller (2025) and the generic image-quality composite of Wei et al. (2025), using mean per-task quality metrics that decouple quality from the number of scheduled tasks. This formulation enables the systematic characterization of quality–coverage trade-offs across five solvers spanning greedy heuristics, single-objective GA, and NSGA-III variants on four scale classes (200 scenarios).

2. Related Work

We examine three lines of research that our study connects: satellite scheduling (2.1), SAR physical quality modeling (2.2), and multi-objective evolutionary optimization (2.3).

2.1. Satellite Scheduling

Satellite scheduling has been studied extensively in the Earth-observation domain. The survey by Ferrari et al. (2025) catalogs over 200 publications spanning exact methods, heuristics, metaheuristics, and learning-based approaches. Exact formulations based on mixed-integer linear programming (MILP) achieve optimality on small instances but fail to scale to operational problem sizes, motivating a broad family of heuristic and metaheuristic alternatives including genetic algorithms, simulated annealing, and tabu search (Wang et al., 2019; Chen et al., 2019). The earliest formal treatments of agile satellite scheduling by Lemaître et al. (2002) and Bianchessi et al. (2007) established the foundational problem structure of time-dependent transition times between observations. Recent work further generalizes these methods to agile satellites, where the freedom to choose observation start times within each visibility window is exploited to pack more observations into a single orbit (Peng et al., 2019; Chen et al., 2024). A parallel line of research has pursued learning-based and hybrid approaches for satellite scheduling. Liu et al. (2024) and Zhou et al. (2025) apply deep reinforcement learning to agile optical satellite scheduling, while Jacquet et al. (2024) combine graph neural networks with Monte Carlo tree search for the same domain. Wang et al. (2026) propose a Lagrangian relaxation-based heuristic for multi-satellite mission planning. Caleiras et al. (2026) formulate super-agile satellite scheduling as a constraint programming problem, capturing the time-dependent transition constraints characteristic of super-agile platforms. Herrmann and Schaub (2023) apply RL to on-board AEOS scheduling, maximizing target collection reward without imaging quality as an objective. Wu et al. (2026) address high-target-density AEOS scheduling with an improved whale optimization algorithm, reporting convergence challenges that echo the density-driven constraint tightening we analyze in § 6.3. Despite their methodological diversity, most prior works treat quality as a fixed property determined at observation-window generation. In contrast, SAR imaging quality varies continuously with acquisition geometry and is directly optimizable within the scheduling window. Chang et al. (2022) propose an early MOEA for SAR satellite scheduling using two adaptive NSGA-II variants on a Gaofen-12 X-band SAR platform; their bi-objective formulation (profit + energy consumption) represents the closest prior work to ours, but it omits imaging quality dimensions. Wei et al. (2025) introduce a neural-policy-guided MOEA that incorporates a composite image quality score as an optimization objective. Kramer and Miller (2025) schedule SAR constellation observations with a hybrid metaheuristic using a weighted visibility–performance composite, embedding a SAR-quality proxy in the scheduler, and Yao et al. (2024) develop a bilevel evolutionary algorithm for multi-satellite scheduling that separates task allocation from detailed observation planning. The common limitation of this prior work is that SAR physical quality metrics—noise-equivalent sigma zero (NESZ), azimuth resolution, ground-range resolution—are either absent or approximated by coarse proxies (e.g., penalizing extreme off-nadir angles). To our knowledge, no existing SAR scheduling method treats continuous, physics-derived quality functions as decomposed optimization objectives in their own right: the closest approaches embed aggregated quality proxies rather than separate geometric and radiometric objectives.

2.2. SAR Physical Quality Modeling

The SAR remote sensing literature provides well-established analytical models linking observation geometry to image quality. Curlander and McDonough (1991), Cumming and Wong (2005), and Moreira et al. (2025) give detailed treatments of SAR physics, including the two quality dimensions relevant to scheduling: geometric resolution and radiometric sensitivity. Geometric resolution decomposes into ground-range resolution $\rho_{\mathrm{rg}} = c/(2B \sin\theta)$, where c is the speed of light, B the radar bandwidth, and θ the incidence angle, and azimuth resolution $\rho_{\mathrm{az}} = D/(2\cos\psi_{\mathrm{sq}})$, where D is the antenna length and ψ_{sq} the squint angle. In operational SAR, amplitude weighting widens ρ_{az} by 10–20% (Curlander and McDonough, 1991), but since this is platform-constant, the unweighted form preserves the relative ordering for optimization. The two resolution components respond oppositely to departure from the zero-Doppler geometry ($\psi_{\mathrm{sq}} = 0$, θ minimal): ρ_{az} worsens as $|\psi_{\mathrm{sq}}|$ grows, whereas ρ_{rg} is coarsest at the minimal incidence of closest approach and improves as θ rises away from it—the origin of the θ-axis trade-off formalized in § 3.2. Radiometric sensitivity, measured by the Noise-Equivalent Sigma Zero (NESZ), deteriorates with the cube of slant range: NESZ $\propto R^3$ for Stripmap SAR after azimuth compression gain (Curlander and McDonough, 1991; Moreira et al., 2025). We adopt a 3D geometric approximation that decomposes the slant range as $R \approx R_{\mathrm{elev}}/\cos\psi_{\mathrm{sq}}$ (where R_{elev} is the elevation-plane range at zero squint), yielding NESZ $\propto 1/(\cos^3\theta \cdot \cos^3\psi_{\mathrm{sq}})$. The $\cos^3\psi_{\mathrm{sq}}$ factor enters through the slant-range decomposition. In the low-squint regime ($|\psi_{\mathrm{sq}}| < 10°$) that dominates the solver-selected schedules reported below, $\cos^3(10°) \approx 0.955$, so the factor's contribution to f_3 variation is at the 4–5% level. The proportionality NESZ $\propto R^3$ captures the dominant geometric dependence; platform-constant terms (noise figure, transmit power, processing gain) do not affect the relative ordering of observation geometries. This cubic dependence means even modest deviations from the optimal viewing geometry make fine observation timing critical for radiometric performance. These quality models are routinely used in SAR system design but their use in scheduling remains limited to post hoc validation of fixed observation plans (Kramer and Miller, 2024) or binary feasibility filtering at the observation-window level (Li et al., 2026). The scheduling literature and the SAR quality modeling literature have developed in parallel: schedulers treat quality as a constraint, while SAR engineers compute quality after the schedule is fixed. Our study connects these two domains by embedding continuous quality functions directly into the optimization objective space.

2.3. Multi-Objective Evolutionary Optimization

Multi-objective evolutionary algorithms (MOEAs) are a standard toolkit for problems with competing objectives. The NSGA-III algorithm (Deb and Jain, 2014) extends the NSGA-II framework to many-objective problems through reference-point-based diversity preservation. The hypervolume (HV) indicator is widely used for comparing Pareto front approximations, but its sensitivity to frontier cardinality creates interpretational challenges when mixing single-objective and multi-objective solvers, a methodological issue we address in § 6.4. To our knowledge, MOEAs have not been applied to SAR scheduling with physical quality objectives; the closest work uses single-objective optimization with post hoc quality evaluation. Our NSGA-III formulation makes these trade-offs explicit by placing coverage profit, geometric quality, and radiometric quality in a unified multi-objective framework.

3. Problem Formulation

3.1. System Model

We consider a single agile SAR satellite in a circular orbit at altitude H, operating in Stripmap mode, which provides the widest continuous swath with uniform quality and analytically tractable NESZ and resolution expressions (Curlander and McDonough, 1991). A problem instance $\mathcal{I} = (\mathcal{T}, \mathcal{P}, \mathcal{O})$ consists of a set of N ground targets $\mathcal{T}$, a satellite platform $\mathcal{P}$, and a SAR imaging model $\mathcal{O}$.

Task Set.. Each task $i \in \{1, \ldots, N\}$ is specified by its geographic coordinates $\mathbf{r}_i = (\phi_i, \lambda_i)$, a priority weight $p_i \in \mathbb{R}^+$, a minimum required ground resolution ρ_i^{req}, a visibility window $\mathcal{W}_i = [a_i, b_i]$ during which the satellite can observe the target, and a fixed observation duration d_i.

Satellite Platform.. The satellite is characterized by its maximum slew rate $\omega_{\max}$ (a conservative single-axis bound on three-axis attitude maneuvers), settling time τ_{settle} after each maneuver, per-orbit energy budget $E_{\max}$ and memory budget $M_{\max}$, and per-observation energy e_i and memory m_i consumption.

SAR Observables.. Given an observation time t_i, the satellite-to-target line-of-sight (LOS) vector $\mathbf{l}(t_i)$ in the Local-Vertical Local-Horizontal (LVLH) frame is fully determined by the satellite's orbital state. From $\mathbf{l}(t_i)$ we derive four geometric quantities deterministically: slant range $R = |\mathbf{l}|$, off-nadir angle $\xi = \arctan 2(\sqrt{l_x^2 + l_y^2},\, l_z)$, incidence angle θ (via the Earth-curvature-corrected mapping $\theta = \arcsin\left(\frac{R_e+H}{R_e}\sin\xi\right)$ where R_e is Earth's mean radius), and squint angle $\psi_{\text{sq}} = \arcsin(|l_x|/R)$ (LVLH x-axis along-track, z-axis nadir-pointing), defined as the angular deviation of the LOS vector from the zero-Doppler plane (Curlander and McDonough, 1991). A critical property is that *all* quality-relevant geometry is a deterministic function of the single decision t_i.

Decision Variables.. A schedule consists of a task selection vector $\mathbf{x} \in \{0, 1\}^N$, an ordered sequence $S = (s_1, \ldots, s_m)$ of the m selected tasks, and a vector of observation start times $\mathbf{t} = (t_{s_1}, \ldots, t_{s_m})$. All geometric parameters (θ_i, $\psi_{\text{sq},i}$, R_i) are *derived* from t_i rather than independent decision variables.

3.2. Objective Functions

We formulate a three-objective optimization problem that captures the tension between scheduling throughput and SAR-specific imaging quality. All quality objectives use the *mean* over scheduled tasks, making them independent of the number of scheduled tasks $m = |\mathcal{S}|$ and thereby producing a genuine "quantity versus quality" Pareto trade-off.

Objective 1: Coverage Profit.. The normalized coverage profit (Eq. 1) is defined relative to the greedy baseline

$$f_1^* = \frac{1}{f_1^{\text{G-BL}}} \sum_{i \in \mathcal{S}} p_i \tag{1}$$

where $f_1^{\text{G-BL}}$ is the profit achieved by the profit-only greedy scheduler (G-BL, § 4.2) on the same scenario. Normalization by G-BL rather than GA-P-BL is conservative for the MOEA coverage deficit: GA-P-BL achieves $f_1^* > 1.0$ at every scale (1.09–1.28, Table 3), so G-BL-normalized f_1^* values understate the coverage gap relative to the best single-objective solver.

Objective 2: Geometric Image Quality.. The ground pixel area of a SAR image is the product of ground-range resolution $\rho_{\text{rg}} = c/(2B\sin\theta)$ and azimuth resolution $\rho_{\mathrm{az}} = D/(2\cos\psi_{\mathrm{sq}})$. We define a dimensionless geometric quality index:

$$f_2 = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \sin\theta_i \cdot \cos\psi_{\mathrm{sq},i} \tag{2}$$

Higher f_2 indicates smaller ground pixel area (better geometric resolution).

Objective 3: NESZ Radiometric Quality.. Noise-Equivalent Sigma Zero (NESZ) measures the minimum detectable backscatter. For Stripmap SAR, NESZ degrades with the cube of slant range, NESZ $\propto R^3 \propto 1/(\cos^3\theta \cdot \cos^3\psi_{\mathrm{sq}})$, after accounting for azimuth compression gain (Curlander and McDonough, 1991). We define the radiometric quality index as:

$$f_3 = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \cos^3\theta_i \cdot \cos^3\psi_{\mathrm{sq},i} \tag{3}$$

Higher f_3 indicates lower NESZ (better radiometric sensitivity).

Physical Trade-Off.. The two quality objectives partition SAR imaging performance by physical mechanism: f_2 captures spatial resolution (geometric), f_3 captures signal-to-noise ratio (radiometric). Their optima are not coincident; f_3 peaks at the zero-Doppler point (θ minimal, $\psi_{\mathrm{sq}} = 0$), whereas f_2 benefits from moderately higher θ because $\sin\theta$ grows with incidence angle. This creates a genuine trade-off: pursuing the cleanest NESZ (zero-Doppler observation) forces acceptance of stretched ground-range pixels; pursuing optimal ground pixel area requires deviating from the zero-Doppler point and accepting elevated radiometric noise. The squint angle ψ_{sq} appears in both f_2 and f_3 (Eqs. 2–3): reducing ψ_{sq} simultaneously improves both resolution components through the $\cos\psi_{\mathrm{sq}}$ factor. In practice, the zero-squint point coincides with the time of closest approach, which minimizes the slant range and, for a spherical Earth, approximately minimizes the incidence angle θ and thus $\sin\theta$; reducing ψ_{sq} therefore lowers f_2 through a geometric (not algebraic) coupling between the two angles, creating a two-axis $[\theta, \psi_{\mathrm{sq}}]$ trade-off structure that limits multi-objective gain once solvers converge to low-squint regimes (Eq. 6 in § 6.4 and its axis decomposition for the exact partition into definitional and scheduler-induced components).

Because θ and ψ_{sq} are derived from the same LOS vector (§ 3.1), the per-task f_2 and f_3 contributions share geometric structure; its quantitative characterization is presented in § 6.4.

3.3. Constraints

Observation-window membership is treated as the domain of each start-time variable rather than as a separate numbered constraint:

$$t_i \in [a_i, b_i - d_i] \quad \forall i \in \mathcal{S}. \tag{4}$$

The remaining four constraints couple task selection, acquisition geometry, temporal feasibility, and resource use. The incidence-angle and squint-angle constraints (C1) are enforced during visibility-window generation; C2–C4 are checked inline by every solver.

C1: Incidence and Squint Angle Constraints.. Each selected acquisition must satisfy the platform incidence-angle range, $\theta^{\min} \le |\theta_i| \le \theta^{\max}$, and the squint-angle limit $|\psi_{\mathrm{sq},i}| \le \psi_{\mathrm{sq}}^{\max}$, where $\psi_{\mathrm{sq}}^{\max}$ is the satellite's maximum achievable squint (typically 45°). Both are observation-geometry limits applied before scheduling.

C2: Attitude Maneuver and Non-Overlap.. Between consecutive observations, the satellite must rotate through the angular separation of their LOS vectors. Let $\mathbf{l}_k$ and $\mathbf{l}_{k+1}$ be the LVLH-frame LOS vectors at t_{s_k} and $t_{s_{k+1}}$. The angular gap $\Delta\eta_k = \arccos(\mathbf{l}_k \cdot \mathbf{l}_{k+1}/(|\mathbf{l}_k||\mathbf{l}_{k+1}|))$ must be traversed within the available inter-observation time:

$$t_{s_{k+1}} \ge t_{s_k} + d_{s_k} + \frac{\Delta\eta_k}{\omega_{\max}} + \tau_{\mathrm{settle}} \quad \forall k \in \{1, \dots, m-1\}. \tag{5}$$

This inequality also enforces non-overlapping task-execution intervals. It is the core coupling in our formulation: changing t_{s_k} shifts $\mathbf{l}_k$, affecting both imaging quality (via θ_k and $\psi_{\mathrm{sq},k}$) and transition feasibility (via $\Delta\eta_k$).

The tightness of C2 scales with task density: as N grows, the fraction of task pairs whose required angular transition $\Delta\eta/\omega_{\max}$ exceeds the available inter-observation time increases, tightening the constraint and making C2 the dominant active constraint in dense regimes.

C3–C4: Energy and Memory Budgets.. $\sum_{i\in\mathcal{S}} e_i \le E_{\max}$ and $\sum_{i\in\mathcal{S}} m_i \le M_{\max}$, where $E_{\max} = 10^7$ and $M_{\max} = 10^{11}$ (arbitrary units) are set sufficiently high to be non-binding under all tested scenarios (typical per-task $e_i \sim 10^2$–10^3, $m_i \sim 10^6$–10^7, far below the caps); they are retained for completeness.

4. Methodology

4.1. Solver Overview

Table 1 summarizes the five solvers compared in this study. We describe the solvers in order of increasing sophistication: from greedy heuristics (G-BL, G-SM) through single-objective metaheuristic (GA-P-BL), to multi-objective evolutionary algorithms (MOEA-2, MOEA-3).

Table 1: Solvers: greedy baselines (G-BL, G-SM), single-objective GA with hot-start (GA-P-BL), and NSGA-III variants with two and three objectives (MOEA-2, MOEA-3)

Label	**Type**	**Objectives**	**Hot-Start**
G-BL	Greedy heuristic	max f_1 (profit-only)	n/a
G-SM	Greedy heuristic	max f_1 + squint minimization	n/a
GA-P-BL	Single-objective GA	max f_1 (G-BL seeded)	std = 0.5
MOEA-2	NSGA-III (2-objective)	$f_1^* + f_2$ (geometric quality)	std = 0.5
MOEA-3	NSGA-III (3-objective)	$f_1^* + f_2 + f_3$ (geo.+NESZ)	std = 0.5

Single-objective solvers (G-BL, G-SM, GA-P-BL) optimize only f_1; their f_2 and f_3 values are computed post hoc on the final schedule for comparison with the multi-objective solvers. The five solvers differ primarily in objective-function configuration rather than algorithmic paradigm. This design is intentional: it isolates the effect of adding physical quality objectives, which is the paper's research question. A comparison across algorithmic families (e.g., NSGA-III vs. MOEA/D (decomposition-based) vs. SMS-EMOA (hypervolume-based)) is left to future work (§ 7.3).

4.2. G-BL: Greedy Baseline

G-BL represents the conventional scheduling approach: it makes no use of SAR imaging physics when ordering or selecting tasks. Tasks are sorted by earliest start time t_{start} and greedily inserted into the schedule at their earliest feasible time. Each task is observed at the off-nadir angle pre-stored in its visibility window (the angle yielding maximum elevation, equivalent to a fixed minimum-incidence-angle policy). If the attitude transition constraint (C2) is violated for a candidate task, its start time is delayed within the window until the constraint is satisfied; if no feasible delay exists, the task is dropped. Since G-BL selects observation times solely for feasibility, its f_2 and f_3 values are incidental by-products of the window geometry. We compute these post hoc on the final schedule to serve as the baseline against which physics-aware solvers are compared.

4.3. G-SM: Greedy Squint-Minimized

G-SM shares G-BL's greedy structure (sort by earliest start time) but modifies the observation time selection: within each visibility window, a task is placed at the discretized time step that minimizes squint angle $\psi_{\text{sq}} = \arcsin(|l_x|/R)$. G-SM minimizes squint angle as a computationally cheap proxy for maximizing f_3 (NESZ radiometric quality); the correlation between squint minimization and f_3 improvement is confirmed by the large effect size (Cliff's $\delta = +0.72$, § 6.2). This strategy serves as a physics-aware greedy baseline: it answers the question "what quality improvement can a simple squint-minimization heuristic achieve, without any multi-objective search?" The squint-minimizing time selection incurs a profit penalty because the optimal-squint time may lie later in the window, reducing the remaining time for subsequent tasks (constraint C2). As with G-BL, f_2 and f_3 are post hoc outputs.

4.4. GA-P-BL: Single-Objective GA with Hot-Start

GA-P-BL tests whether a genetic algorithm optimizing only f_1 can outperform greedy scheduling by exploiting temporal flexibility. Each chromosome encodes a vector of normalized start times $\boldsymbol{\tau} \in [0, 1]^N$, where task i's actual start time is $t_i = t_i^{\text{earliest}} + \tau_i \cdot (t_i^{\text{latest}} - d_i - t_i^{\text{earliest}})$. The task selection vector $\mathbf{x}$ emerges from constraint resolution: a task is dropped if no feasible insertion time exists within its window after satisfying C2. The population of 100 individuals is initialized from G-BL's solution with Gaussian noise (std $= 0.5$) added to each τ_i. The GA runs for 200 generations with simulated binary crossover ($p_c = 0.9$, $\eta_c = 20$) and polynomial mutation ($p_m = 0.1$, $\eta_m = 20$). f_2 and f_3 are computed post hoc on the best-found schedule.

4.5. MOEA-2 and MOEA-3: Multi-Objective Evolutionary Algorithms

MOEA-2 and MOEA-3 extend the GA-P-BL chromosome encoding to multi-objective optimization via the NSGA-III framework (Deb and Jain, 2014). Both use a population of 100 individuals over 200 generations with the same genetic operators as GA-P-BL. Reference points are generated via the Das–Dennis systematic approach, producing $H = \binom{n_{\text{obj}}+p-1}{p}$ structured reference directions, where n_{obj} is the number of objectives and $p = 12$ the divisions per objective dimension ($H = 13$ for MOEA-2, $H = 91$ for MOEA-3). The entire population is hot-started from the G-BL solution perturbed by Gaussian noise (std = 0.5), and an at-least-one-task constraint prevents empty solutions from dominating the Pareto front. As every solver is hot-started from the non-empty G-BL solution (§ 4.2), this constraint is inactive in the reported experiments. MOEA-2 optimizes two objectives: f_1^* (normalized profit) and f_2 (mean geometric quality, Eq. 2). MOEA-3 adds f_3 (mean NESZ radiometric quality, Eq. 3) as a third objective. The comparison between MOEA-2 and MOEA-3 directly tests whether adding the NESZ objective (f_3) produces Pareto-front differentiation beyond what f_2 alone captures, or whether the θ/ψ_{sq} geometric coupling renders the third objective redundant in the single-pass, single-satellite setting.

5. Experimental Setup

5.1. Scenario Design

Table 2 defines the four scenario classes.

Table 2: Scenario classes: four density levels (S1–S4) with $N = 20, 100, 300, 500$ targets, 50 scenarios per class

Class	Targets	Look Direction	Description
S1	20	Both	Small, low-density
S2	100	Both	Medium, moderate density
S3	300	Both	Medium-large, transition region
S4	500	Both	Large, high-density

All scenarios share a common satellite platform abstracted as a generic C-band Stripmap SAR with bilateral-looking capability. The platform follows a circular sun-synchronous orbit at 693 km altitude (inclination $\approx 97.8°$, LTAN 06:00, dawn–dusk) and carries a C-band Stripmap instrument with antenna length $D = 12.3$ m and bandwidth $B = 56.5$ MHz. It accesses both left- and right-side incidence angles in the range $[18°, 47°]$ (signed convention: $\theta \in [-47°, -18°] \cup [18°, 47°]$). This generic bilateral configuration is selected to exercise the full incidence–squint trade-off space.

Targets are distributed along the satellite ground track within the accessible swath (cross-track half-width $\pm 5.5°$ from nadir, roughly ± 608 km), with three distribution types per scenario class: uniform random, clustered (5 clusters), and mixed (half uniform, half clustered). The five scenarios within each class correspond to distribution type and random-seed variations; the 50 scenarios per class therefore consist of 5 distribution–seed combinations (labeled A–E in the scenario files) $\times$ 10 random seeds each. The four density classes (S1–S4) span sparse reconnaissance ($N = 20$) to dense monitoring ($N = 500$). Other intermediate densities ($N = 50, 400$) are omitted to limit computational cost. To directly constrain the transition midpoint, two additional intermediate classes—S7 ($N = 150$) and S8 ($N = 200$), 50 scenarios each generated under the same Sentinel-1 bilateral configuration—were run for MOEA-2 and the two greedy baselines (G-BL, G-SM) and are used solely in the scale-sensitivity analysis in § 6.3; they do not enter the five-solver pooled comparisons.

Each task is assigned a priority weight p_i drawn uniformly from the integer set $\{1, 2, \ldots, 10\}$, and a fixed observation duration $d_i = 30\,\mathrm{s}$. Visibility windows are computed via orbit propagation over two orbital periods ($\sim$200 min), yielding 1–2 windows per target on average, each spanning several minutes. Windows are filtered by incidence-angle and minimum-elevation (5° above the local horizon) constraints. The resulting windows are approximately evenly split between left- and right-looking opportunities ($\sim$50% each).

5.2. Solver Configuration

All five solvers run on a single CPU core to ensure a uniform computational environment. G-BL and G-SM are deterministic greedy heuristics with no tunable hyperparameters. GA-P-BL and the MOEA variants use the population size, generation budget, and genetic operators described in §4.4 and §4.5, with the entire MOEA population hot-started from the G-BL solution. We verified that increasing the population to 200 does not qualitatively change the results.

5.3. Evaluation Metrics

We use hypervolume (HV) as the primary quality indicator for multi-objective comparison. Statistical significance is assessed via the Friedman test for overall differences across 5 solvers $\times$ 200 scenarios, followed by pairwise Wilcoxon signed-rank comparisons with Bonferroni correction ($\alpha = 0.05/10 = 0.005$). Per-objective and ablation comparisons are reported with raw p-values alongside the Bonferroni threshold. Effect sizes are measured by Cliff's δ, classified by the Vargha–Delaney convention ($|\delta| < 0.147$ negligible,

< 0.33 small, < 0.474 medium, ≥ 0.474 large). Hypervolume is computed with reference point $(0, 0, 0)$ in the per-solver-pooled min–max-normalized objective space. Per-solver pooling ensures that each solver's HV reflects its own objective range, but it also means that the normalization basis differs across solvers; cross-solver HV comparisons should therefore be interpreted as relative rankings within a shared normalization framework rather than absolute cardinal differences. All-solver-pooled normalization was not used because the single-objective solvers (G-BL, G-SM, GA-P-BL) produce only one point each, which would collapse the min–max range for their objectives.

5.4. Experiment Design Overview

This study comprises three classes of experiments. The first is the main experiment, running all five solvers (G-BL, G-SM, GA-P-BL, MOEA-2, MOEA-3) on 50 scenarios per density class (S1–S4), totaling 200 scenarios. Results from the main experiment are presented in §6.1–§6.4 and §6.2. The second is the ablation experiment, which successively removes the squint component (variant B), the incidence-angle component (variant C), and all physical objectives (variant D) from the MOEA-3 framework, running each variant on the same 200 scenarios (§6.5). The third comprises robustness control experiments—hot-start dependence, perturbation-sensitivity sweep, random-search calibration, and search-budget control— which verify that the main and ablation findings are not artifacts of initialization or parameter choices (§6.6).

6. Results and Analysis

6.1. Overall Solver Performance

Table 3 reports all five solvers across all four scenario classes. The full-scale view shows why the value of multi-objective optimization cannot be judged from S4 alone. At S1–S2, MOEA-2 improves f_2 relative to G-BL while accepting a substantial coverage reduction; at S3–S4, the quality gain contracts and MOEA coverage approaches the G-BL reference. Across every scale, GA-P-BL achieves the highest coverage while preserving geometric quality close to G-BL. G-SM occupies a different operating regime, exchanging coverage and f_2 for substantially higher f_3.

Table 3: Solver performance across all scenario classes (mean ± standard deviation over 50 scenarios per class). Higher values are preferred for f_1^*, f_2, and f_3.

Class	**Solver**	f_1^*	f_2	f_3
S1 ($N = 20$)	G-BL	1.00 ± 0.00	0.541 ± 0.068	0.325 ± 0.174
	G-SM	0.85 ± 0.16	0.474 ± 0.077	0.435 ± 0.158
	GA-P-BL	1.09 ± 0.14	0.550 ± 0.069	0.294 ± 0.178
	MOEA-2	0.68 ± 0.29	0.585 ± 0.055	0.240 ± 0.153
	MOEA-3	0.66 ± 0.30	0.569 ± 0.066	0.288 ± 0.151
S2 ($N = 100$)	G-BL	1.00 ± 0.00	0.540 ± 0.059	0.305 ± 0.148
	G-SM	0.74 ± 0.09	0.454 ± 0.059	0.456 ± 0.118
	GA-P-BL	1.28 ± 0.22	0.548 ± 0.056	0.276 ± 0.144
	MOEA-2	0.82 ± 0.25	0.564 ± 0.044	0.235 ± 0.129
	MOEA-3	0.83 ± 0.24	0.561 ± 0.044	0.245 ± 0.128
S3 ($N = 300$)	G-BL	1.00 ± 0.00	0.578 ± 0.025	0.217 ± 0.071
	G-SM	0.64 ± 0.07	0.501 ± 0.034	0.361 ± 0.073
	GA-P-BL	1.15 ± 0.15	0.579 ± 0.024	0.203 ± 0.072
	MOEA-2	0.98 ± 0.04	0.587 ± 0.021	0.175 ± 0.065
	MOEA-3	0.97 ± 0.03	0.587 ± 0.021	0.175 ± 0.066
S4 ($N = 500$)	G-BL	1.00 ± 0.00	0.580 ± 0.025	0.210 ± 0.071
	G-SM	0.61 ± 0.09	0.506 ± 0.035	0.356 ± 0.075
	GA-P-BL	1.17 ± 0.27	0.579 ± 0.025	0.203 ± 0.073
	MOEA-2	0.98 ± 0.03	0.589 ± 0.021	0.169 ± 0.066
	MOEA-3	0.97 ± 0.04	0.589 ± 0.021	0.170 ± 0.067

Table 4: Hypervolume across 200 scenarios for the two multi-objective solvers, with G-BL as a single-point reference (single-point solvers G-SM and GA-P-BL are omitted from this comparison; see § 6.4). HV comparisons between single-point and multi-point solvers are confounded by point-count asymmetry (see text).

Solver	**HV**	**SD**
MOEA-3	0.173	0.045
MOEA-2	0.169	0.037
G-BL	0.140	0.017

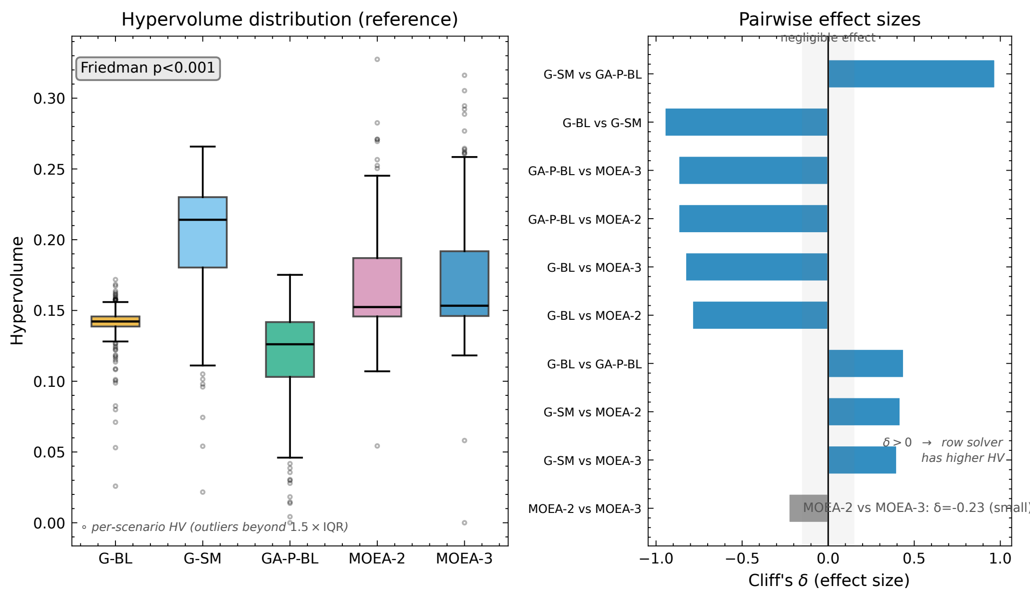

Figure 1: Statistical validation: (left) hypervolume distribution; (right) pairwise Cliff's δ effect sizes. The gray band marks negligible effects ($|\delta| < 0.147$).

Figure 1 statistically validates these solver comparisons through the hypervolume distribution and pairwise Cliff's δ effect sizes.

The following sections deepen this overall analysis from five perspectives: §6.2 examines the special trade-off of squint-aware scheduling, §6.3 investigates scale sensitivity (how multi-objective advantage varies with target density), §6.4 reveals the geometric mechanism of Pareto compression, §6.5 verifies the condition dependence of this mechanism through ablation, and §6.6 separates initialization and search-budget confounds through robustness controls.

6.2. Squint-Aware Special Regime

The squint-minimized greedy solver (G-SM) improves NESZ radiometric quality (f_3) relative to the profit-only baseline (G-BL) by selecting observation times that minimize squint angle within each visibility window. Across all 200 scenarios, this produces a large positive effect on f_3 (Cliff's $\delta = +0.72$, $p = 1.7 \times 10^{-34}$) at a small aggregate cost in coverage (Cliff's $\delta = -0.25$, $p = 7.0 \times 10^{-34}$, on f_1^*; raw f_1 is used here because G-BL yields constant $f_1^* = 1.0$ by definition, for which the normalized-profit comparison degenerates to a one-sample test against a fixed value). The trade-off, however, is strongly density-dependent. In sparse scenarios (S1, $N = 20$), G-SM retains $f_1^* = 0.85 \pm 0.16$ (-15% vs. G-BL at 1.00) while improving f_3 from 0.32 ± 0.17 to 0.44 ± 0.16 ($+34\%$). In dense scenarios (S4, $N = 500$), the f_1 cost escalates: G-SM achieves $f_1^* = 0.61 \pm 0.09$ (-39%), with f_3 rising from 0.210 ± 0.071 to 0.356 ± 0.075 ($+70\%$). Figure 2 visualizes this density-dependent trade-off: the Locally Estimated Scatterplot Smoothing (LOESS) trend in panel (c) identifies a transition region around $N = 100$ where the f_1 penalty grows rapidly. In dense regimes, the squint-minimization strategy becomes costly because tightly packed time windows offer fewer alternative observation times at low squint.

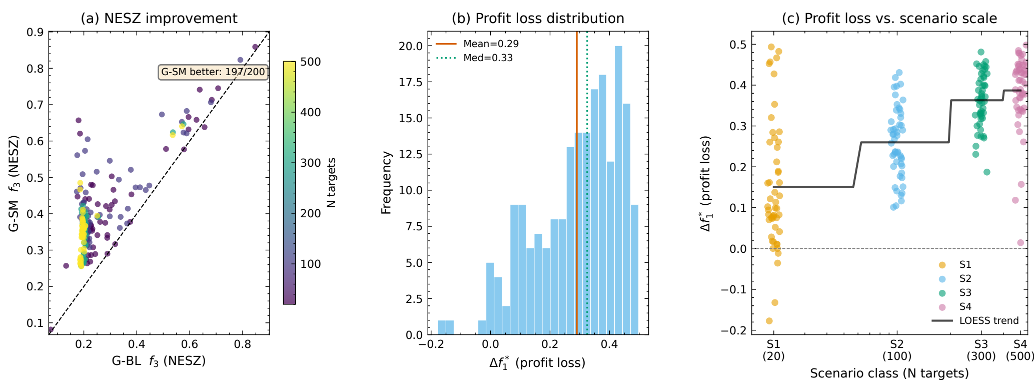

Figure 2: Effect of squint-aware scheduling: (a) f_3 scatter; (b) Δf_1^* histogram; (c) Δf_1^* vs. N with LOESS trend

6.3. Scale Sensitivity

The gap between MOEA-2 and G-BL varies systematically with scenario density (Fig. 3, Table 5). MOEA-2's coverage deficit relative to G-BL narrows with increasing scale: from $f_1^* = 0.68 \pm 0.29$ at S1 ($N = 20$) to 0.82 ± 0.25 at S2 ($N = 100$), 0.94 ± 0.06 at S7 ($N = 150$), 0.96 ± 0.04 at S8 ($N = 200$), 0.98 ± 0.04 at S3 ($N = 300$), and 0.98 ± 0.03 at S4 ($N = 500$). The f_2 improvement over G-BL follows the same contraction, falling from $+8.2\%$ (S1) to $+4.4\%$ (S2) and to near zero at the intermediate densities ($+0.4\%$ at S7, $+0.2\%$ at S8), before a slight rebound to $+1.6\%$ (S3) and $+1.5\%$ (S4). The proportion of scenarios where MOEA-2's coverage falls measurably short of G-BL ($f_1^* < 0.95$, i.e., a coverage deficit exceeding 5%) falls from 88% (S1) through 62% (S2), 42% (S7), and 30% (S8) to 16% (S3), with a slight rise to 24% (S4). The deficit is monotonic from S1 to S3; the rise from S3 (16%)

to S4 (24%) is not statistically significant ($\chi^2(1) = 1.00$, $p = 0.32$). Between $N = 100$ and $N = 300$, the multi-objective solver transitions from actively trading off coverage for quality to operating near the greedy baseline.

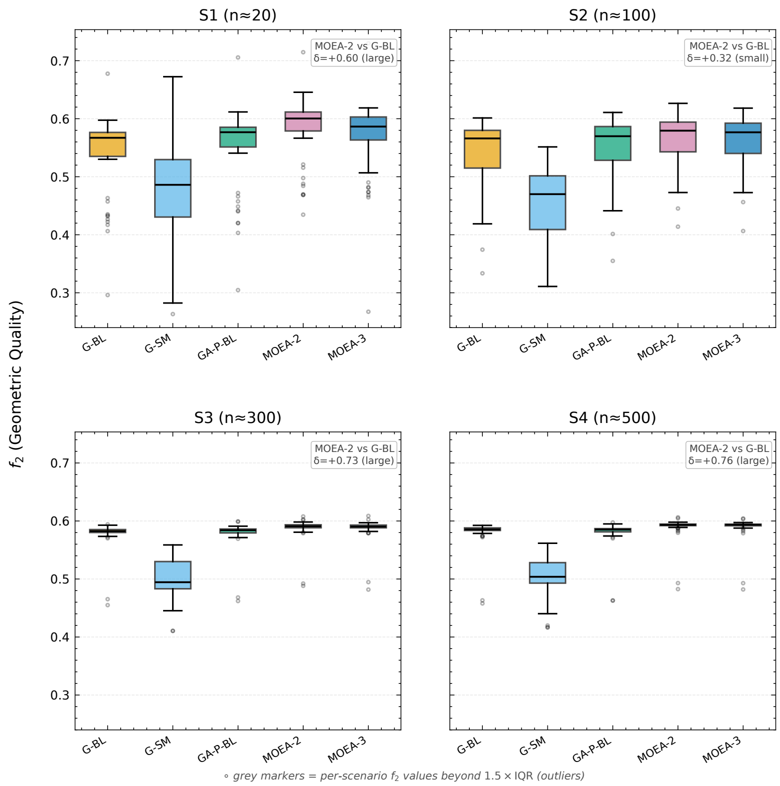

Figure 3: Geometric quality (f_2) by scenario group and solver

Table 5: Scale sensitivity of MOEA-2 vs. G-BL

Class	N	f_1^* (MOEA-2)	f_2 improvement	% $f_1^* < 0.95$
S1	20	0.68 ± 0.29	+8.2%	88%
S2	100	0.82 ± 0.25	+4.4%	62%
S7	150	0.94 ± 0.06	+0.4%	42%
S8	200	0.96 ± 0.04	+0.2%	30%
S3	300	0.98 ± 0.04	+1.6%	16%
S4	500	0.98 ± 0.03	+1.5%	24%

Fitting a logistic regression to the proportion of scenarios with $f_1^* < 0.95$ across the six density points ($N = 20, 100, 150, 200, 300, 500$) yields a transition midpoint $N_{50} \approx 140$ targets (95% CI $[117, 169]$ via bootstrap with 1000 resamples). The two intermediate densities ($N = 150, 200$) constrain the transition directly, narrowing the CI from its four-point width $[111, 227]$ to $[117, 169]$. Note that the deficit proportion is non-monotonic at the high-density end (S3: 16% → S4: 24%, $\chi^2(1) = 1.00$, $p = 0.32$); N_{50} should therefore be read as a coarse transition indicator rather than a precise threshold. By $N \approx 140$, half the scenarios show MOEA-2 within 5% of G-BL coverage, and beyond $N \approx 300$ the deficit proportion stabilizes near 16–24%. The transition region $N = 100$–300 therefore captures the density range where the multi-objective advantage transitions from measurable to negligible for most scenarios.

6.4. Mechanism of Pareto Compression

Phenomenon 1: Front compression. Fig. 4 visualizes the Pareto front on a representative S4 scenario. The front contains 76 ± 17 MOEA-3 non-dominated solutions (MOEA-2: 78 ± 20) but is compressed: it spans roughly 0.3 in f_1^* and only 0.06 in f_2, against a total span of $\Delta f_2 \approx 0.10$ in sparse S1 scenarios, confirming the scale-dependent saturation documented in §6.3. In dense regimes, solver convergence to low-squint geometries combined with schedule saturation limits quality differentiation regardless of solver strategy. The MOEA-2 and MOEA-3 fronts are nearly coincident, indicating that adding f_3 as a third objective produces no additional differentiation in this setting.

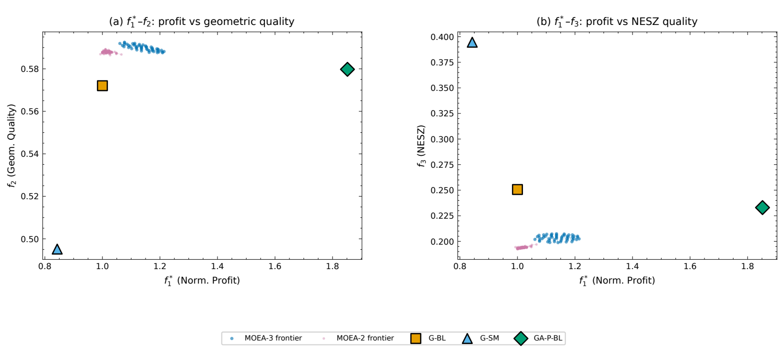

Figure 4: Pareto front on a representative S4 scenario ($N = 500$): MOEA-3 frontier with MOEA-2 overlay and baseline solver markers. Panel (a): f_1^*–f_2; panel (b): f_1^*–f_3.

The f_1^*–f_2 projection (panel a) shows a narrow but nonzero trade-off surface: the frontier spans from full-coverage solutions ($f_1^* \approx 1.0$, $f_2 \approx 0.56$) to quality-oriented solutions ($f_1^* \approx 0.7$, $f_2 \approx 0.62$). Overlaying the single-point baseline solutions shows that G-BL sits at the "full-coverage, moderate-quality" corner of the frontier, while G-SM occupies a qualitatively different region ($f_1^* \approx 0.6$, $f_3 \approx 0.35$) that lies outside the MOEA-3 Pareto surface, a consequence of G-SM's aggressive squint minimization.

Phenomenon 2: Three-objective redundancy. The near-identity of MOEA-2 and MOEA-3 fronts under the NSGA-III framework is quantified by pairwise Cliff's $\delta = +0.132$ (f_2, negligible, $p = 3.9\times10^{-10}$) and $\delta = -0.176$ (f_3, small, $p = 5.6 \times 10^{-14}$).

The HV ranking (Table 4) differs from the coverage ranking (Table 3): among the three tabulated solvers, G-BL ranks first in coverage (f_1^*) but third in HV, while the NSGA-III variants rank first and second in HV (MOEA-3, then MOEA-2), despite their lower coverage ranking. This reversal is driven by point-count asymmetry: single-objective solvers contribute one point each to the HV reference set, while multi-objective solvers contribute 76–78 non-dominated points (SD 17–20). HV therefore conflates front quality with front cardinality, and the ranking should not be interpreted as a standalone quality judgment. The core phenomenon—front compression—is independent of HV ranking.

Mechanism: geometric coupling and low-squint convergence.

Because θ and ψ_{sq} are not independent but both derived from the same LOS vector $\mathbf{l} = (l_x, l_y, l_z)$ (§ 3.1), the per-task f_2 and f_3 contributions share geometric structure. To separate definitional coupling from scheduler-induced correlation, we compute the analytical null distribution by treating θ and ψ_{sq} as independently and uniformly distributed within the accessible envelope ($\theta \in [15°, 50°]$, $|\psi_{\text{sq}}| \leq 45°$, slightly wider than the C1 constraint $[18°, 47°]$ to provide a conservative null baseline) and evaluating the moment integrals of $f_{2,i} = \sin\theta_i \cos\psi_{\text{sq},i}$ and $f_{3,i} = \cos^3\theta_i \cos^3\psi_{\text{sq},i}$ in closed form (validated by Monte Carlo, $r_{\text{MC}} = -0.5120$, $N_{\text{MC}} = 2\times10^5$). The null correlation is

$$r_{\text{null}} = \frac{\text{Cov}(f_2, f_3)}{\sqrt{\text{Var}(f_2)\,\text{Var}(f_3)}} \;=\; -0.51, \tag{6}$$

where the independence of θ and ψ_{sq} under the null factorizes the moments, e.g., $E[f_2 f_3] = E[\sin\theta\cos^3\theta]\,E[\cos^4\psi_{\text{sq}}]$. The *negative* sign exposes a structural conflict, not redundancy: in the θ-only limit (fixed squint), $f_{2,i} \propto \sin\theta$

and $f_{3,i} \propto \cos^3\theta$ are strictly anti-monotonic ($r_\theta = -0.997$), since f_2 favors high incidence while f_3 favors low incidence; in the ψ_{sq}-only limit (fixed θ), both depend on $\cos\psi_{\text{sq}}$ positively ($r_\psi = +0.997$), since low squint improves both. Under uniform geometry the θ-conflict dominates, yielding $r_{\text{null}} < 0$.

Empirical correlation and low-squint convergence. Pooled across all scheduled tasks from the 200 scenarios, the per-task empirical correlation between f_2 and f_3 within the solver-selected schedules is negative but weaker than the envelope: for the full-physics MOEA-3 (variant A) it ranges from -0.69 (S1) to -0.17 (S4), and for the no-physics variant D from -0.44 (S1) to -0.18 (S4), decreasing in magnitude as scenario density rises (Table 6). The θ-conflict therefore persists inside the schedules rather than being reversed; scheduled observations are not drawn uniformly from the envelope, and all solvers, including the greedy baselines, select low-squint geometries that suppress the θ-conflict and align f_2 with f_3 along the shared $\cos\psi_{\text{sq}}$ factor, but the residual negative correlation corresponds to the θ-axis conflict space where genuine f_2–f_3 trade-offs remain. The 3-objective redundancy is therefore an empirical convergence effect—not a mathematical identity between f_2 and f_3 but a consequence of solver convergence to low-squint regimes that narrows, without eliminating, the incidence-angle conflict.

Envelope versus schedule correlation. The contrast between envelope correlation and schedule correlation is stark. Across C1-feasible visibility samples ($|\psi_{\text{sq}}| \leq 45°$, computed across all visibility windows from 20 scenarios (5 per density class S1–S4, the five smallest random seeds per class), approximately 3.9×10^5 window–time-step pairs at 10 s resolution), $\text{corr}(f_2, f_3) = -0.95$, a strong negative correlation dominated by the incidence-angle conflict. This envelope correlation of -0.95 differs from the factorized null $r_{\text{null}} = -0.51$ (§ 6.4) because the latter assumes independent θ and ψ_{sq}, whereas real visibility geometry couples the two angles through the LOS vector, making the θ-conflict dominate. The solver-selected schedules thus move the correlation toward zero—from -0.95 in the visibility envelope to ≈ -0.69 to -0.17 within the full-physics schedules—but do not reverse its sign: the incidence-angle conflict is narrowed by low-squint convergence, not eliminated. This partial attenuation is an active effect of the solvers, not a consequence of geometric envelope constraints: envelope constraints alone would leave the correlation at -0.95. Ablation studies independently support this interpretation. With all physical objectives removed (variant D, § 6.5), the schedule-level correlation at S1 is $r \approx -0.44$—closer to zero than under the full-physics variant A ($r \approx -0.69$)—and both variants converge to the same level at high density ($r \approx -0.18$ at S4). In the sparse regime the physical objectives force the solver to retain solutions that traverse the f_2–f_3 quality trade-off, preserving a stronger negative correlation; without them, the solver maximizes coverage alone, and temporal packing incidentally favors low-squint windows, attenuating the residual correlation further. At high density this structural packing dominates irrespective of the objective formulation, so the ablation contrast vanishes; it is largest where the quality trade-off is active (S1), not where scheduling structure alone enforces low-squint convergence. Consistent with this scale-dependent saturation, the per-density hypervolume contracts at higher density: at S4 the pooled HV is 0.72 (MOEA-2) and 0.67 (MOEA-3) of its S1 value, i.e. the S4/S1 HV ratio is below unity, indicating a more compact front rather than degradation (the front still spans $\Delta f_1^* \approx 0.3$). We caution that the Pareto fronts reported here are best-known approximations obtained via NSGA-III; whether they represent the true Pareto-optimal set is unknown without exact methods, which remain intractable for $N \geq 20$ in this problem class.

6.5. Physical-Objective Ablation

To quantify the contribution of incidence-angle (θ) and squint-angle (ψ_{sq}) modeling to solver performance, we repeat the MOEA-3 experiment with four objective-function variants:

A. Full physical (baseline) $f_2 = \frac{1}{|\mathcal{S}|} \sum \sin\theta_i \cdot \cos\psi_{\text{sq},i}$, $f_3 = \frac{1}{|\mathcal{S}|} \sum \cos^3\theta_i \cdot \cos^3\psi_{\text{sq},i}$ (identical to the standard configuration in § 6.4).

B. No squint $f_2 = \frac{1}{|\mathcal{S}|} \sum \sin\theta_i$, $f_3 = \frac{1}{|\mathcal{S}|} \sum \cos^3\theta_i$ (removes ψ_{sq} component).

C. No incidence $f_2 = \frac{1}{|\mathcal{S}|} \sum \cos\psi_{\text{sq},i}$, $f_3 = \frac{1}{|\mathcal{S}|} \sum \cos^3\psi_{\text{sq},i}$ (removes θ component).

D. No physics $f_2 = f_3 = 1$ (constant per-task objective; solver receives no geometric information; f_2, f_3 shown in Table 6 are post hoc diagnostics computed from the actual observation geometry of the resulting schedule). Variant D serves as a null-model control: it tests whether the NSGA-III solver's performance is attributable to the physical content of f_2 and f_3, or merely to the presence of additional objectives that structure the search.

All four variants use identical NSGA-III parameters (population = 100, generations = 200, Das-Dennis $p = 12$), the same 200 scenario files, and the same G-BL hot-start injection. Only the objective functions differ, making this a controlled ablation rather than a retuning experiment. Within-solver ablation with identical NSGA-III parameters and the same G-BL seed injection controls confounding factors sufficiently that the Wilcoxon signed-rank test is appropriate for paired comparison of objective-function variants. Each variant completes 4 scenario classes × 50 seeds = 200 runs (total: 800 runs across A–D). Degradation is reported as $\Delta\% = (\mathrm{A} - \mathrm{V})/\mathrm{A} \times 100$, where A is the full physical baseline and V is the variant; negative values indicate the variant outperforms the baseline on f_1^*.

Table 6: Per-class ablation results. Bold: $p < 0.005$ (Wilcoxon signed-rank vs. A; consistent with the Friedman–Wilcoxon framework of §5.3).

Class	**V**	f_1^*	f_2	f_3	f_1/n	$n_{\mathbf{sel}}$	$\Delta f_1^*\%$
S1 ($N = 20$)	A	0.66 ± 0.30	0.569	0.288	5.93 ± 1.11	11.1 ± 4.5	(baseline)
	B	0.73 ± 0.34	0.565	0.281	5.57 ± 1.06	12.8 ± 5.6	**−9.5%**
	C	0.52 ± 0.22	0.575	0.300	5.91 ± 1.26	8.8 ± 3.5	**+21.6%**
	D	0.81 ± 0.34	0.565	0.266	5.40 ± 1.05	14.7 ± 5.7	**−22.1%**
S2 ($N = 100$)	A	0.83 ± 0.24	0.561	0.245	5.56 ± 0.53	47.6 ± 21.6	(baseline)
	B	0.83 ± 0.24	0.560	0.244	5.55 ± 0.56	47.8 ± 21.5	−0.3%
	C	0.80 ± 0.21	0.561	0.247	5.44 ± 0.49	46.6 ± 20.2	+3.2%
	D	0.84 ± 0.25	0.561	0.243	5.46 ± 0.60	48.6 ± 21.3	−1.2%
S3 ($N = 300$)	A	0.97 ± 0.03	0.587	0.175	5.50 ± 0.23	151.1 ± 39.1	(baseline)
	B	0.98 ± 0.05	0.587	0.175	5.52 ± 0.23	151.7 ± 38.9	−0.9%
	C	0.98 ± 0.03	0.587	0.175	5.52 ± 0.23	151.1 ± 38.7	−0.4%
	D	0.98 ± 0.05	0.587	0.175	5.51 ± 0.23	151.1 ± 39.2	−0.2%
S4 ($N = 500$)	A	0.97 ± 0.04	0.589	0.170	5.51 ± 0.20	190.2 ± 56.8	(baseline)
	B	0.98 ± 0.06	0.589	0.169	5.50 ± 0.20	190.6 ± 57.0	−0.1%
	C	0.98 ± 0.05	0.589	0.170	5.51 ± 0.21	191.0 ± 56.4	−0.6%
	D	0.98 ± 0.07	0.589	0.169	5.50 ± 0.19	190.3 ± 56.2	−0.2%

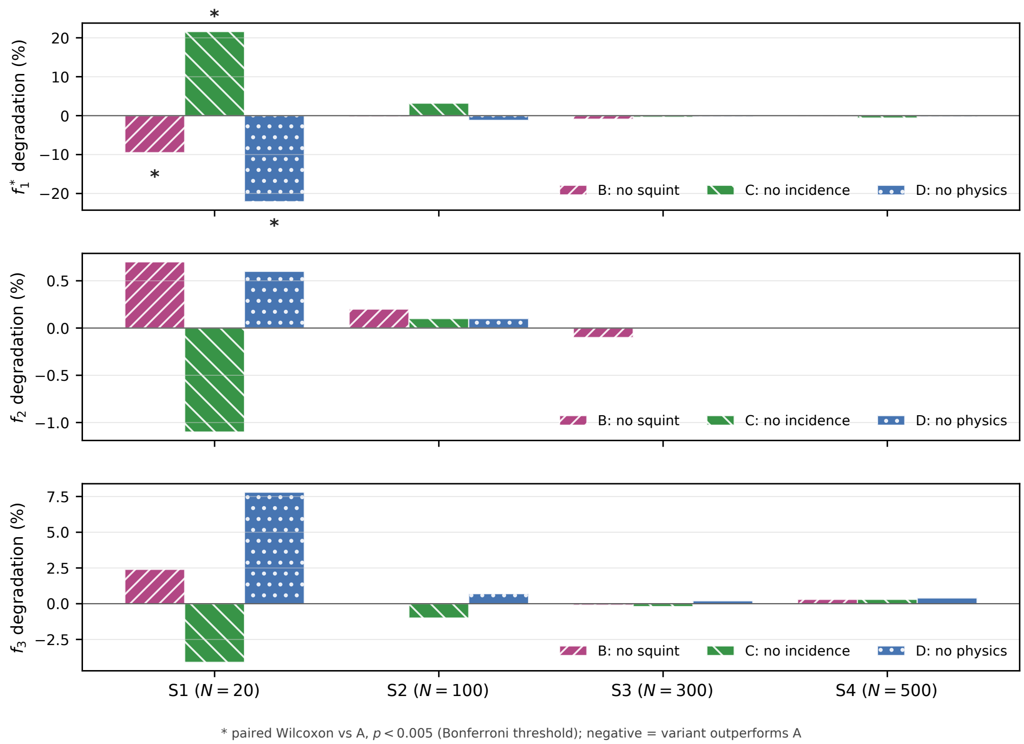

Figure 5: Ablation results across objective-function variants B–D and scenario classes S1–S4. Bars show degradation relative to the full-physics baseline A for normalized profit f_1^* and post hoc geometric-quality metrics f_2 and f_3; negative values indicate that a variant outperforms A. Asterisks mark paired Wilcoxon comparisons against A with $p < 0.005$ (Bonferroni threshold).

The ablation results (Table 6, Fig. 5) reveal that the influence of physical objective modeling is concentrated in sparse scenarios and vanishes as scenario density increases — a pattern consistent with our scale-sensitivity analysis (§ 6.3).

At S1 ($N = 20$), all three ablations differ significantly from baseline A at the manuscript's $p < 0.005$ threshold. Variant C (no incidence) degrades f_1^* by +21.6% ($p = 6.2 \times 10^{-6}$). Conversely, Variant D (no physics) improves apparent f_1^* by −22.1% ($p = 3.3 \times 10^{-7}$) by selecting about one-third more tasks (14.7 vs. 11.1), while Variant B (no squint) yields a −9.5% change ($p = 1.2 \times 10^{-3}$). Thus physical objectives materially alter solver behavior in the sparse regime, although the direction of the coverage change depends on which component is removed.

The asymmetry between variants B and C at S1 is informative. Variant C (no incidence) degrades f_1^* by +21.6% yet produces *higher* post hoc f_2 and f_3 than baseline A (0.575 vs. 0.569 and 0.300 vs. 0.288). Removing θ leaves both objectives as functions of ψ_{sq} alone ($\cos\psi_{\text{sq}}$ and $\cos^3\psi_{\text{sq}}$), both maximized at zero squint. Variant C therefore behaves like a near-G-SM strategy: it aggressively minimizes squint at the expense of coverage, trading profit for radiometric quality. Variant B (no squint), by contrast, retains incidence-angle variation and achieves the opposite trade-off—improved coverage ($\Delta f_1^* = -9.5\%$, i.e., 9.5% above baseline A) at negligible quality cost. This asymmetry confirms that the coverage–quality trade-off is driven primarily by the θ axis, consistent with the dominance of incidence-angle conflict in the envelope correlation ($r \approx -0.95$, § 6.4).

The counterintuitive result that Variant D (no physics) outperforms baseline A on f_1^* by 22.1% at S1 is visible in the task counts: D selects 14.7 tasks versus A's 11.1, yet the post hoc geometric quality of the two schedules is nearly identical ($f_2 = 0.565$ vs. 0.569, a difference of $< 1\%$). This near-identity of geometric quality despite the absence of physical objectives reinforces the geometric-coupling interpretation: the low-squint convergence documented in § 6.4 is partly a structural property of dense observation-window geometry, but the per-task correlation analysis shows that physical-objective guidance attenuates the incidence-angle conflict further in the sparse regime ($r \approx -0.69$ for A versus -0.44 for D at S1), an effect that vanishes as density increases. The coverage gap between A and D arises from the mechanics of NSGA-III in three-objective space: the Pareto front maintained by A contains solutions with fewer tasks but higher per-task quality, which can dominate solutions with more tasks but lower per-task quality in the crowding-distance sorting used by NSGA-III's environmental selection. Variant D, with constant $f_2 = f_3 = 1$, effectively collapses to single-objective profit maximization within the NSGA-III diversity machinery, and therefore selects tasks more aggressively. This structural explanation is consistent with the search-budget control (§ 6.6), where doubling population and generations does not separate A from D.

At S2 ($N = 100$), none of the three normalized-f_1^* comparisons reaches $p < 0.005$. All variants lie within 3.2% of baseline A, and their per-task profit (5.44–5.56) and selected-task counts (46.6–48.6) are similar. The physical-objective signal is already much weaker than at S1.

At S3–S4 ($N \geq 300$), every variant converges to within 1% of the baseline for f_1^* and to within 0.5% for f_2, f_3, and n_{sel}. The per-task quality metrics

(f_1/n) are identical to two decimal places (5.50–5.52) across all variants and scene classes. The f_2 value is locked at 0.587–0.589, varying by less than 0.3%. This numerical invariance is the direct signature of the scale-dependent saturation discussed in § 6.3: once the density of candidate observation windows saturates the schedule (N exceeds the maximum selectable targets under the attitude maneuver and squint constraints), the objective function only penalizes infeasible solutions rather than differentiating among feasible ones. In this regime, the NSGA-III solver acts as a constraint-filtered profit maximizer regardless of whether f_2 and f_3 carry geometric meaning (A) or serve as constant placeholders (D). Search-budget and random-initialization controls for this interpretation are reported in § 6.6.

6.6. Robustness and Calibration

The mechanism above must be separated from initialization and search-budget effects. We therefore use four control experiments.

Hot-start dependence.. On 10 S1, 4 S2, 5 S3, and 5 S4 scenarios (3 seeds each), random-initialized MOEA-2 yields f_1^* deficits of approximately -0.18 ± 0.26, -0.42 ± 0.13, -0.46 ± 0.04, and -0.33 ± 0.04 (mean $\pm$ SD across runs) relative to hot-start MOEA-2, respectively. G-BL seeding therefore materially improves coverage convergence at every tested density; notably, the deficit is largest in the dense S3 class rather than the sparsest one, indicating that high target density does not by itself make random initialization sufficient. This constrains the coverage-ceiling claim: proximity to G-BL is initialization-dependent. By contrast, random-init solutions attain slightly higher f_2 (0.605–0.624 vs. 0.594–0.603 for the hot start), so the f_2–f_3 geometric relationship is not created by the anchor. The coverage ceiling is therefore a G-BL initialization effect rather than an artifact of the MOEA framework.

Perturbation sensitivity.. A sweep over $\sigma \in \{0.1, 0.3, 0.5, 0.7\}$ on MOEA-3 (3 seeds per value, 10 S3 scenarios per seed) changes mean f_2 by < 0.002 and mean f_3 by < 0.001. The between-perturbation correlation of mean f_2 and f_3 across σ levels remains near zero ($r \in [-0.003, 0.176]$; distinct from the per-task f_2–f_3 correlation within schedules reported in § 6.4), while coverage declines by only about 3% from $\sigma = 0.1$ to 0.7. Redundancy therefore persists across $\sigma = 0.1$–0.7 and is not a byproduct of the $\sigma = 0.5$ perturbation level.

Random-search calibration.. On five S1 scenarios, 5,000 random task selections per scenario produce mean $f_1^* = 0.54 \pm 0.01$ and a 90th percentile of 0.96 ± 0.02; the best sample reaches 1.03 ± 0.03. High coverage under random sampling occurs only by chance; the MOEA's gains therefore reflect structured search rather than luck.

Search-budget control.. Doubling population and generation budget to 200 and 400 on five S3 and five S4 scenarios does not separate full-physics A from no-physics D: $\Delta f_1^* = -0.005$ at S3 and $+0.002$ at S4, with post hoc $f_2 \approx 0.595$ for both. A random-init D control (15 S3 and 15 S4 runs) also converges to low squint ($f_2 = 0.605 \pm 0.003$). Since doubling the budget leaves A–D indistinguishable, the invariance is not a budget artifact. These controls collectively favor the low-squint convergence explanation over hot-start residue, noise-level sensitivity, random chance, or insufficient budget, while not ruling out all algorithm-specific effects.

7. Discussion

7.1. Summary of Findings

We define a solver as *Pareto-sufficient* at scale N if no alternative solver in the comparison set achieves both higher f_1^* and higher f_2 (or f_3) with statistical significance ($p < 0.005$, Cliff's $|\delta| > 0.33$). This operational criterion separates cases where multi-objective optimization yields a measurable quality–coverage advantage from cases where it does not. By this criterion, MOEA-2 is Pareto-sufficient at sparse-to-moderate densities (its pooled f_2 advantage over G-BL is large: Cliff's $\delta = +0.55$, $p < 10^{-33}$), whereas at $N \geq 300$ the scale-sensitivity (§ 6.3) and ablation (§ 6.5) results show no solver retaining a joint significant advantage, so the greedy baseline is operationally sufficient. Our experiment across 200 systematically varied scenarios yields five interconnected findings: **(1)** the quality advantage of multi-objective optimization saturates with target density (+8.2% at $N = 20$ falling to +1.5% at $N = 500$); **(2)** the three-objective formulation is effectively redundant, because per-task f_2–f_3 correlation within schedules stays negative but weakens with density (≈ -0.69 to -0.17 for the full-physics solver), confirming that low-squint convergence narrows, rather than removes, the incidence-angle conflict; **(3)** G-SM occupies a distinct high-radiometric-quality regime at substantial density-dependent coverage cost; **(4)** solver outcomes become invariant to the choice of physical objectives at $N \geq 300$ (ablation study);

and **(5)** coverage convergence toward the G-BL anchor is conditional on the hot start, not a structural property. The first four reflect geometric properties of the single-pass observation geometry (finding (2) being an empirical convergence effect under those properties, § 6.4); the fifth reflects solver initialization.

The Friedman test confirms overall differences among the five solvers across all 200 scenarios ($\chi^2 = 506.56$, $p \approx 10^{-108}$). Pairwise Wilcoxon signed-rank tests use the Bonferroni threshold $\alpha = 0.005$; interpretation relies on Cliff's δ given the large sample size. MOEA-2 and MOEA-3 differ statistically but only weakly in practical terms ($p = 1.4 \times 10^{-4}$, $\delta = -0.23$ on HV), whereas contrasts involving G-SM are large because it occupies an extreme radiometric-quality regime.

Independently of the multi-objective question, GA-P-BL achieves the highest coverage at every scale through single-objective temporal search (Table 3), delimiting the present claims to the marginal value of adding physical objectives rather than to profit-oriented search alone.

7.2. Operational Implications

The scale dependence has direct consequences for mission planning. In sparse regimes ($N \leq 100$), MOEA-2 achieves an 8.2% quality improvement at S1 but at a steep coverage penalty ($f_1^* = 0.68$ at $N = 20$, i.e., 32% profit loss relative to G-BL). Whether this trade-off is acceptable depends on the application: for maritime surveillance tasks where a 4–8% resolution improvement (~ 0.3 m at C-band: $\rho_{\text{rg}} \approx 5.3$ m at $\theta = 30°$) may affect whether adjacent targets within the same pixel are resolved or remain merged at baseline resolution (Curlander and McDonough, 1991), the coverage cost may be justified; for area-monitoring missions, the greedy baseline remains preferable. In dense regimes ($N \geq 300$), the multi-objective advantage falls below 2% improvement in f_2, while G-SM offers the highest radiometric quality (+70% f_3 at S4) at a 39% coverage cost; this gain comes from the one-line squint-minimization heuristic rather than from multi-objective search. The logistic midpoint $N_{50} \approx 140$ (§ 6.3) provides a concrete transition estimate between these regimes. Our results suggest a conditional operational guideline: for sparse observation requests ($N \lesssim 100$), MOEA-2 may provide geometric quality gains that outweigh the coverage penalty in specific contexts; for dense monitoring campaigns ($N \gtrsim 300$), greedy methods (G-SM for radiometric priority, G-BL for coverage priority) approach Pareto-optimal coverage and may be preferable. However, relative to the strongest coverage-only solver (GA-P-BL), the MOEA-2 coverage gap persists at 13–14% even at $N \geq 300$ (paired per-scenario comparison; vs. 2% vs. G-BL), because G-BL is a weaker reference point.

7.3. Limitations

Our study is bounded by scope decisions that define directions for future work. The 3-objective redundancy reported above is specific to the single-pass, single-satellite Stripmap setting: it reflects solver convergence to low-squint regimes that suppress the θ-axis conflict (§ 6.4, Eq. 6), not a mathematical identity between f_2 and f_3. In multi-pass or multi-satellite settings with greater angular diversity, the θ-conflict would re-emerge and the third objective may become discriminative. Similarly, ScanSAR, Spotlight, and TOPS modes introduce different NESZ and resolution dependencies beyond Stripmap, and the scenario design uses a fixed orbit (693 km circular LEO) without exploring sensitivity to orbital parameters. The hot-start control experiment (§ 6.6) indicates that G-BL initialization materially improves MOEA convergence; the coverage convergence reported here is conditional on this initialization, and full-scale random-init experiments are needed to establish the unconditional behavior. The hot-start dependence implies that the coverage-ceiling result reflects the combination of G-BL prior knowledge with MOEA search; whether the MOEA can independently discover high-coverage solutions without greedy seeding remains an open question. Similarly, the NSGA-III crowding-distance explanation for the S1 variant-D reversal (§ 6.5) awaits verification with alternative crowding operators. Finally, we do not benchmark against learning-based schedulers (e.g., Chen et al. (2025)), which represent an alternative methodological family; a systematic comparison of physics-aware heuristic methods against data-driven approaches is left to future work.

8. Conclusion

We described a physics-aware multi-objective optimization approach for agile SAR satellite scheduling that integrates NESZ radiometric quality and geometric resolution as explicit optimization objectives alongside coverage profit. Five solvers (spanning greedy heuristics, single-objective evolutionary search, and NSGA-III with two and three objectives) were compared on 200 systematically varied scenarios across four density classes. The central finding is a scale-dependent empirical saturation of the marginal benefit of physical objectives in G-BL-seeded MOEA formulations for single-pass, single-satellite Stripmap SAR operations. The saturation arises from solver convergence to low-squint regimes under which the geometric coupling aligns f_2 and f_3 along the shared $\cos \psi_{\mathrm{sq}}$ factor, rendering the three-objective formulation effectively redundant. Coverage convergence toward the G-BL anchor is conditional on the hot start; the geometric redundancy is the anchor-independent result (§ 6.4). GA-P-BL still exceeds the anchor through single-objective temporal search, so this result delimits the benefit of adding physical objectives. This saturation boundary applies to the evaluated target-density range (N = 20–500). For mission planning, the operational implication is conditional: multi-objective optimization provides measurable benefit only when observation requests are sparse; at dense monitoring cadences, greedy methods approach Pareto-optimal coverage. These findings establish an empirical saturation boundary for what physics-aware multi-objective scheduling can contribute to single-satellite SAR operations, and provide a baseline against which multi-satellite, multi-pass, and multi-mode extensions can be measured.

AI and AI-Assisted Technologies

During the preparation of this work the author used AI-assisted tools for manuscript preparation only. The author reviewed all AI-assisted output and takes full responsibility for the content of this article.

Acknowledgments

The author thanks Prof. LIU Fanming (Harbin Engineering University) for guidance and support throughout this research. Prof. Liu is a co-author of Wu et al. (2026); his contribution to the present work was limited to advisory discussions on research direction and did not include experimental design, data analysis, or manuscript preparation. This work was supported by no specific external grant; it was performed as part of the author's doctoral studies at Harbin Engineering University.

CRediT Authorship Contribution Statement

Liu Shuhao: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing.

Data Availability Statement

Solver code, scenario generators, fixed seeds and configurations, processed JSON/CSV results, analysis scripts, and figures are available from the corresponding author. The companion repository is publicly available at `https://github.com/liushuhao/sar-quality-aware-scheduling`. Generated binary scenario files and per-run pickle objects are not tracked; they can be regenerated using the supplied scripts.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The work was performed as part of the author's doctoral studies at Harbin Engineering University, College of Intelligent Systems Science and Engineering.

References

Bianchessi, N., Cordeau, J.F., Desrosiers, J., Laporte, G., Raymond, V., 2007. A heuristic for the multi-satellite, multi-orbit and multi-user management of Earth observation satellites. European Journal of Operational Research 177, 750–762. doi:`10.1016/j.ejor.2005.12.026`.

Caleiras, M., Moniz, S., Nascimento, P.J., 2026. A constraint programming model for the super-agile earth observation satellite imaging scheduling problem, in: Proc. Int. Conf. on Operations Research and Enterprise Systems (ICORES). `arXiv:2601.11967`. preprint; ICORES 2026 proceedings, to appear.

Chang, Z., Zhou, Z., Li, R., 2022. Observation scheduling for a state-of-the-art sar earth observation satellite: Two adaptive multi-objective evolutionary algorithms. Computers & Industrial Engineering 169. doi:10.1016/j.cie.2022.108252.

Chen, M., Du, Y., Tang, K., Xing, L., Chen, Y., Chen, Y., 2024. Learning to construct a solution for the agile satellite scheduling problem with time-dependent transition times. IEEE Transactions on Systems, Man, and Cybernetics: Systems 54, 5991–6002. doi:10.1109/tsmc.2024.3411640.

Chen, M., Wei, L., Chun, J., He, L., Xiang, S., Xing, L., Chen, Y., 2025. A neural priority model for agile earth observation satellite scheduling using deep reinforcement learning. Applied Soft Computing 174. doi:10.1016/j.asoc.2025.112984.

Chen, X., Reinelt, G., Dai, G., Spitz, A., 2019. A mixed integer linear programming model for multi-satellite scheduling. European Journal of Operational Research 275, 419–434. doi:10.1016/j.ejor.2018.11.058.

Cumming, I., Wong, F., 2005. Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation. Artech House.

Curlander, J., McDonough, R., 1991. Synthetic Aperture Radar: Systems and Signal Processing. Wiley.

Deb, K., Jain, H., 2014. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: Solving problems with box constraints. IEEE Transactions on Evolutionary Computation 18, 577–601. doi:10.1109/TEVC.2013.2281535.

Ferrari, B., Cordeau, J.F., Delorme, M., Iori, M., Orosei, R., 2025. Satellite scheduling problems: A survey of applications in earth and outer space observation. Computers & Operations Research 173. doi:10.1016/j.cor.2024.106875.

Herrmann, A., Schaub, H., 2023. Reinforcement learning for the agile earth-observing satellite scheduling problem. IEEE Transactions on Aerospace and Electronic Systems , 1–13doi:10.1109/taes.2023.3251307.

Jacquet, A., Infantes, G., Benazera, E., Roussel, S., Meuleau, N., Baudoui, V., Guerra, J., 2024. Earth observation satellite scheduling with graph neural networks and monte carlo tree search, in: SPAICE2024 / IWPSS 2025. arXiv:2408.15041.

Kramer, E., Miller, D., 2024. Visibility metric for planning synthetic aperture radar observations in challenging terrain. IEEE Transactions on Geoscience and Remote Sensing 62, 1–14. doi:10.1109/tgrs.2024.3487050.

Kramer, E., Miller, D., 2025. Scheduling observations of visibility stressing terrain with SAR satellite constellations. Acta Astronautica 235, 405–420. doi:10.1016/j.actaastro.2025.05.017.

Lemaître, M., Verfaillie, G., Jouhaud, F., Lachiver, J.M., Bataille, N., 2002. Selecting and scheduling observations of agile satellites. Aerospace Science and Technology 6, 367–381. doi:10.1016/S1270-9638(02)01173-2.

Li, L., Feng, Q., Li, K., Liang, Y., 2026. Efficient generation and quality screening of visible windows for regional SAR reconnaissance. arXiv:2604.19539. preprint.

Liu, Z., Xiong, W., Jia, Z., Han, C., 2024. Two-stage deep reinforcement learning method for agile optical satellite scheduling problem. Complex & Intelligent Systems 11, 35. doi:10.1007/s40747-024-01667-x.

Moreira, A., Krieger, G., Villano, M., Younis, M., Prats-Iraola, P., Zink, M., 2025. Spaceborne synthetic aperture radar: Future technologies and mission concepts. Proceedings of the IEEE 114, 183–219. doi:10.1109/jproc.2025.3621586.

Peng, G., Dewil, R., Verbeeck, C., Gunawan, A., 2019. Agile earth observation satellite scheduling: An orienteering problem with time-dependent profits and resources. Computers & Operations Research 111, 84–98. doi:10.1016/j.cor.2019.05.030.

Wang, F., Chen, Y., He, L., Chen, J., 2026. A lagrangian relaxation-based heuristic algorithm for multiple agile earth observation satellite scheduling with time-dependent constraint. European Journal of Operational Research 332, 84–100. doi:10.1016/j.ejor.2025.12.047.

Wang, J., Demeulemeester, E., Hu, X., Qiu, D., Liu, J., 2019. Exact and heuristic scheduling algorithms for multiple earth observation satellites under uncertainties of clouds. IEEE Systems Journal 13, 3556–3567. doi:10.1109/JSYST.2018.2874223.

Wei, L., Cui, Y., Chen, M., Wan, Q., 2025. Multi-objective neural policy approach for agile earth satellite scheduling considering image quality. Swarm and Evolutionary Computation 94. doi:10.1016/j.swevo.2025.101857.

Wu, W., Wu, X., Liu, F., Zhu, J., 2026. Improved whale optimization algorithm for increasing the task planning efficiency of agile earth observation satellites under a high target density. Acta Astronautica 238, 1–11. doi:10.1016/j.actaastro.2025.09.005.

Yao, F., Chen, Y., Wang, L., Chang, Z., 2024. Bilevel evolutionary algorithm for large-scale multiobjective task scheduling in multi-agile earth observation satellite systems. IEEE Transactions on Systems, Man, and Cybernetics: Systems 54, 3512–3524. doi:10.1109/tsmc.2024.3359265.

Zhou, G., Jin, Z., Liu, D., 2025. Deep reinforcement learning-based two-phase hybrid algorithm for multi-agile earth observation satellite scheduling. Remote Sensing 17. doi:10.3390/rs17121972.